

Technology Stack (Architecture & Stack)

Project Design Phase-II – Alternative Version

Technical Architecture

- User Interface: Web-based interface developed using HTML, CSS and JavaScript.
- Application Logic-1: Handles the core application processes using JavaScript.
- Application Logic-2: Manages application interactions and page-level processing using JavaScript.
- Application Logic-3: Provides additional client-side validation and process handling using JavaScript.
- Database: No database is used; the application works with static frontend data.
- Cloud Database: Not applicable for this frontend-only application.
- File Storage: Local project files containing HTML, CSS, JavaScript and required assets.
- External API-1: No external API is required for the application.
- External API-2: No external API is required for the application.
- Machine Learning Model: Not applicable.
- Infrastructure: The application can be executed locally using a web browser.

Table-1: Components & Technologies

S.No	Component	Technology / Description
1	User Interface	HTML, CSS, JavaScript – Web interface
2	Application Logic-1	JavaScript – Core application process
3	Application Logic-2	JavaScript – Interaction handling
4	Application Logic-3	JavaScript – Validation and process handling
5	Database	Not used – Static frontend application
6	Cloud Database	Not applicable
7	File Storage	Local project filesystem
8	External API-1	Not used
9	External API-2	Not used
10	Machine Learning Model	Not applicable
11	Infrastructure	Local system / Web browser

Table-2: Application Characteristics

S.No	Characteristics	Description / Technology
1	Open-Source Frameworks	HTML, CSS and JavaScript
2	Security	Client-side validation and controlled inputs
3	Scalable Architecture	Simple frontend architecture
4	Availability	Runs locally using a web browser
5	Performance	Lightweight HTML, CSS and JavaScript

References

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>